

Elastic Stack SOC L1 Home Lab

We'll build a **Elastic Stack SOC L1 home lab** step by step.

Lab goal:

```
Windows 10/11 VM
      ↓
Elastic Agent
      ↓
Elasticsearch
      ↓
Kibana / Elastic Security
      ↓
Windows security logs
      ↓
Search logs in Kibana
```

We'll use **Windows** as the endpoint because Windows Event Logs are extremely important for SOC L1 jobs.

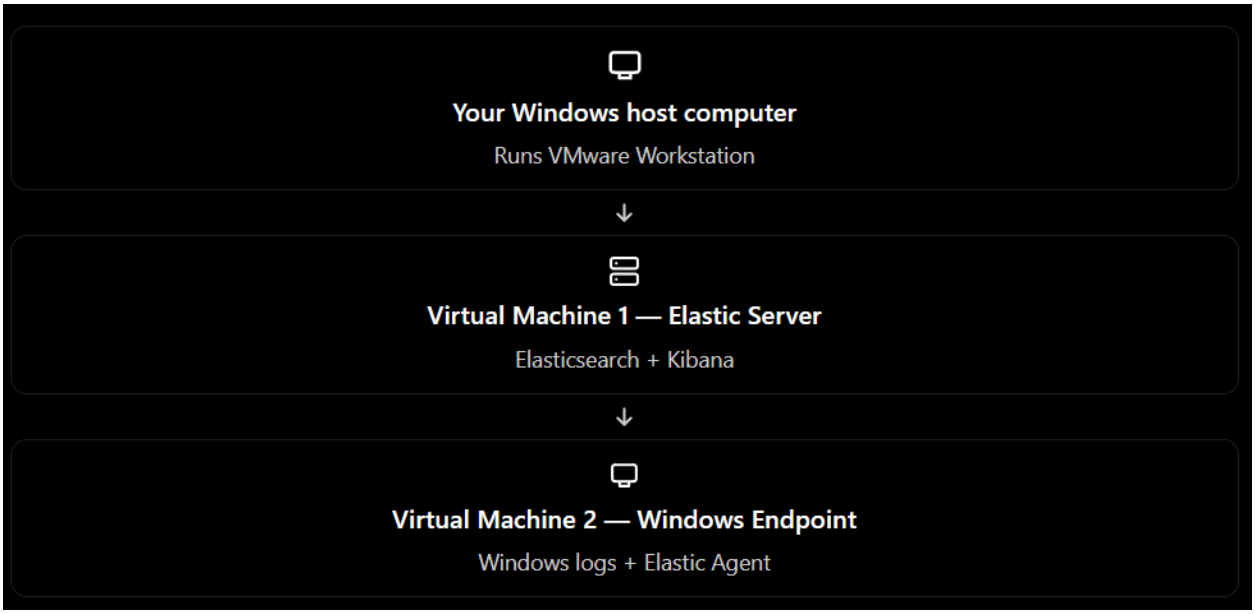
What we'll practice

Stage	What you'll learn
1	Build the lab environment
2	Install Elastic Stack
3	Configure Elastic Agent

Stage	What you'll learn
4	Collect Windows Event Logs
5	Search logs in Kibana

Stage 1 — Set up your SOC lab

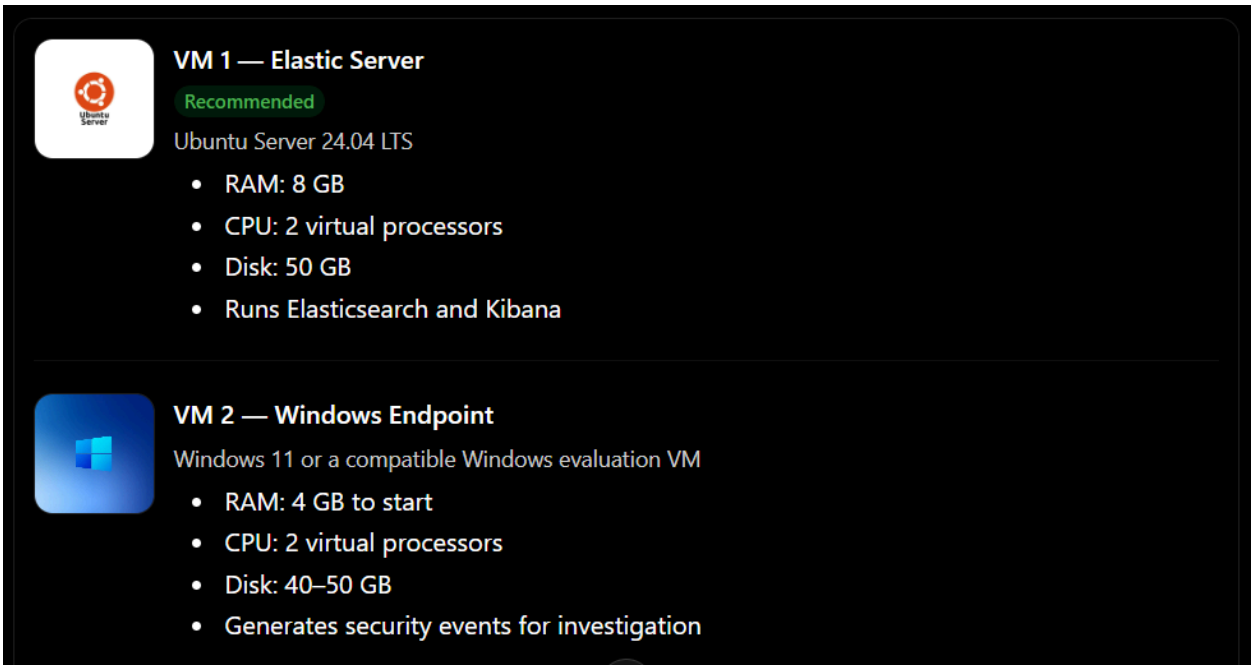
1. Understand what we're building



Windows endpoint will generate security events. Elastic will collect those events, display them in Kibana, and help you detect suspicious activity.

Stage 2 — Prepare your virtual machines

Recommended lab configuration:





Stage 2A — Start with a compatibility check

```
elastic-soc-server@elastic-soc:~$ curl -I https://artifacts.elastic.co
HTTP/2 302
server: nginx/1.29.7
content-length: 0
location: https://www.elastic.co/downloads/
via: 1.1 google
date: Tue, 13 Oct 2026 16:06:57 GMT
cache-control: public, max-age=600
age: 85
x-content-type-options: nosniff
strict-transport-security: max-age=63072000; includeSubDomains; preload
alt-svc: h3=":443"; ma=2592000
```

Step 1 — Prepare the installation tools:

```
elastic-soc-server@elastic-soc:~$ sudo apt update
sudo apt install -y curl gnupg ca-certificates
Hit:1 http://pk.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://pk.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://pk.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [601 kB]
Get:6 http://pk.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [734 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [138 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
```

```
Get:9 http://security.ubuntu.com/ubuntu resolute-security/uni
verse amd64 Components [55.6 kB]
Get:10 http://pk.archive.ubuntu.com/ubuntu resolute-updates/m
ain amd64 Components [110 kB]
```

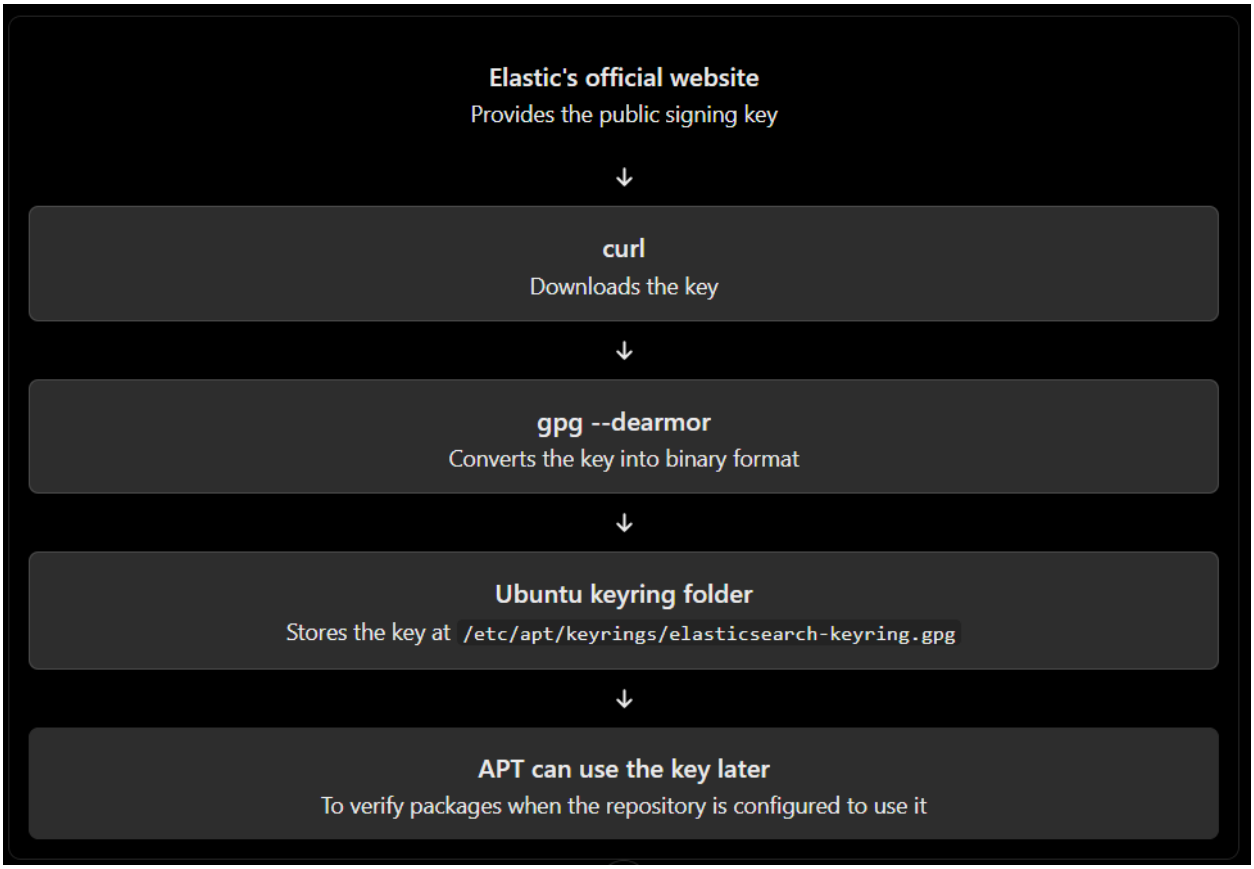
```
Get:11 http://pk.archive.ubuntu.com/ubuntu resolute-updates/u
niverse amd64 Packages [323 kB]
Get:12 http://pk.archive.ubuntu.com/ubuntu resolute-updates/u
niverse amd64 Components [201 kB]
Get:13 http://pk.archive.ubuntu.com/ubuntu resolute-backport
s/universe amd64 Components [1,060 B]
Fetched 2,623 kB in 2s (1,295 kB/s)
```

```
22 packages can be upgraded. Run 'apt list --upgradable' to s
ee them.
curl is already the newest version (8.18.0-1ubuntu2.7).
gnupg is already the newest version (2.4.8-4ubuntu3.1).
gnupg set to manually installed.
ca-certificates is already the newest version (20260601~26.0
4.1).
ca-certificates set to manually installed.
Summary:
```

```
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 22
```

Stage 2B — Add Elastic's official signing key

we'll configure package verification. This lets Ubuntu verify that packages from Elastic are signed with Elastic's trusted key.




```
curl -fsSL https://artifacts.elastic.co/GPG-KEY-elasticsearch
\
| sudo gpg --dearmor --yes -o /etc/apt/keyrings/elasticsearch
-keyring.gpg
```

Then download and store Elastic's public signing key:

```
elastic-soc-server@elastic-soc:~$ curl -fsSL https://artifact
s.elastic.co/GPG-KEY-elasticsearch \
| sudo gpg --dearmor --yes -o /etc/apt/keyrings/elasticsearch
-keyring.gpg
```

This downloads Elastic's public signing key and stores it in the system keyring.

Step 1 — Add the Elastic 9.x repository

```
elastic-soc-server@elastic-soc:~$ echo "deb [signed-by=/etc/a
pt/keyrings/elasticsearch-keyring.gpg] https://artifacts.elas
tic.co/packages/9.x/apt stable main" \
| sudo tee /etc/apt/sources.list.d/elastic-9.x.list
deb [signed-by=/etc/apt/keyrings/elasticsearch-keyring.gpg] h
ttps://artifacts.elastic.co/packages/9.x/apt stable main
```

This tells Ubuntu where to find Elastic 9.x packages.

Step 2 — Refresh the package list

```
elastic-soc-server@elastic-soc:~$ sudo apt update
Hit:1 http://pk.archive.ubuntu.com/ubuntu resolute InRelease
```

```
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InR
elease
```

```
Hit:3 http://pk.archive.ubuntu.com/ubuntu resolute-updates In
Release
```

```
Hit:4 http://pk.archive.ubuntu.com/ubuntu resolute-backports
InRelease
```

```
Get:5 https://artifacts.elastic.co/packages/9.x/apt stable In
Release [3,249 B]
Get:6 https://artifacts.elastic.co/packages/9.x/apt stable/ma
in amd64 Packages [62.1 kB]
Fetched 65.4 kB in 2s (39.0 kB/s)
22 packages can be upgraded. Run 'apt list --upgradable' to s
ee them.
```

Stage 2C — Install Elasticsearch

Elasticsearch is the engine that stores, indexes, and searches your security logs. We'll install it first, then configure Kibana.

1. Install Elasticsearch

```
sudo apt install -y elasticsearch
```

This downloads and installs the Elasticsearch package

2. Enable Elasticsearch at startup

After the installation finishes successfully

```
sudo systemctl daemon-reload
sudo systemctl enable elasticsearch
```

daemon-reload makes systemd reload its service definitions.

enable configures Elasticsearch to start automatically when Ubuntu boots.

```
elastic-soc-server@elastic-soc:~$ sudo systemctl daemon-reloa
d
sudo systemctl enable elasticsearch
Created symlink '/etc/systemd/system/multi-user.target.wants/
elasticsearch.service' → '/usr/lib/systemd/system/elasticsear
ch.service'.
```

3. Start Elasticsearch

```
sudo systemctl start elasticsearch
```

Wait about 30–60 seconds, then check its status:

```
elastic-soc-server@elastic-soc:~$ sudo systemctl status elast
icsearch --no-pager
● elasticsearch.service - Elasticsearch
Loaded: loaded (/usr/lib/systemd/system/elasticsearch.servic
e; enabled; preset: enabled)
Active: active (running) since Tue 2026-10-13 21:44:47 PKT; 1
min 9s ago
```

4. Test Elasticsearch

```
curl --cacert /etc/elasticsearch/certs/http_ca.crt -u elastic
https://localhost:9200
```

Stage 3 — Install Kibana

Kibana is the web interface for Elastic. Instead of using only terminal commands, you'll use Kibana to search security logs, view dashboards, create detection rules, and investigate alerts.

Step 1 — Install Kibana

```
sudo apt install -y kibana
```

This installs Kibana from the same Elastic 9.x repository that provided Elasticsearch 9.5.5.

```
elastic-soc-server@elastic-soc:~$ sudo apt install -y kibana
Installing:
```

```
kibana
```

```
Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 19
Download size: 458 MB
Space needed: 1,405 MB / 55.8 GB available
```

```
Get:1 https://artifacts.elastic.co/packages/9.x/apt stable/ma
in amd64 kibana amd64 9.5.5 [458 MB]
Fetched 458 MB in 2min 2s (3,764 kB/s)
```

```
Selecting previously unselected package kibana.
(Reading database… 170994 files and directories currently ins
talled.)
Preparing to unpack …/kibana_9.5.5_amd64.deb…
Unpacking kibana (9.5.5)…
Setting up kibana (9.5.5)…
Creating kibana group... OK
Creating kibana user... OK
Created Kibana keystore in /etc/kibana/kibana.keystore
```

Step 2 — Enable Kibana at startup

```
sudo systemctl daemon-reload
sudo systemctl enable kibana
```

```
Your Windows 11 computer
|
| Web browser
v
KIBANA :5601
(SOC dashboard)
|
| Secure connection
```

```
v
ELASTICSEARCH :9200
(Stores and searches logs)
```

Step 3 — Create a Kibana enrollment token

```
sudo /usr/share/elasticsearch/bin/elasticsearch-create-enroll
ment-token -s kibana
```

Step 4 — Start Kibana

```
sudo systemctl enable kibana
sudo systemctl start kibana
```

The warning in your log identifies one configuration issue we should fix before setting up Elastic Agent.

```
Kibana is running
|
v
Fleet plugin initializes
|
v
Missing encryption key for saved objects / API keys
|
v
Configure encryption keys and restart Kibana
```

These keys let Kibana securely encrypt sensitive information used by features such as Fleet.

Step 5 — Generate Kibana encryption keys

```
sudo /usr/share/kibana/bin/kibana-encryption-keys generate
```

Step 6 — Add the keys to Kibana's configuration

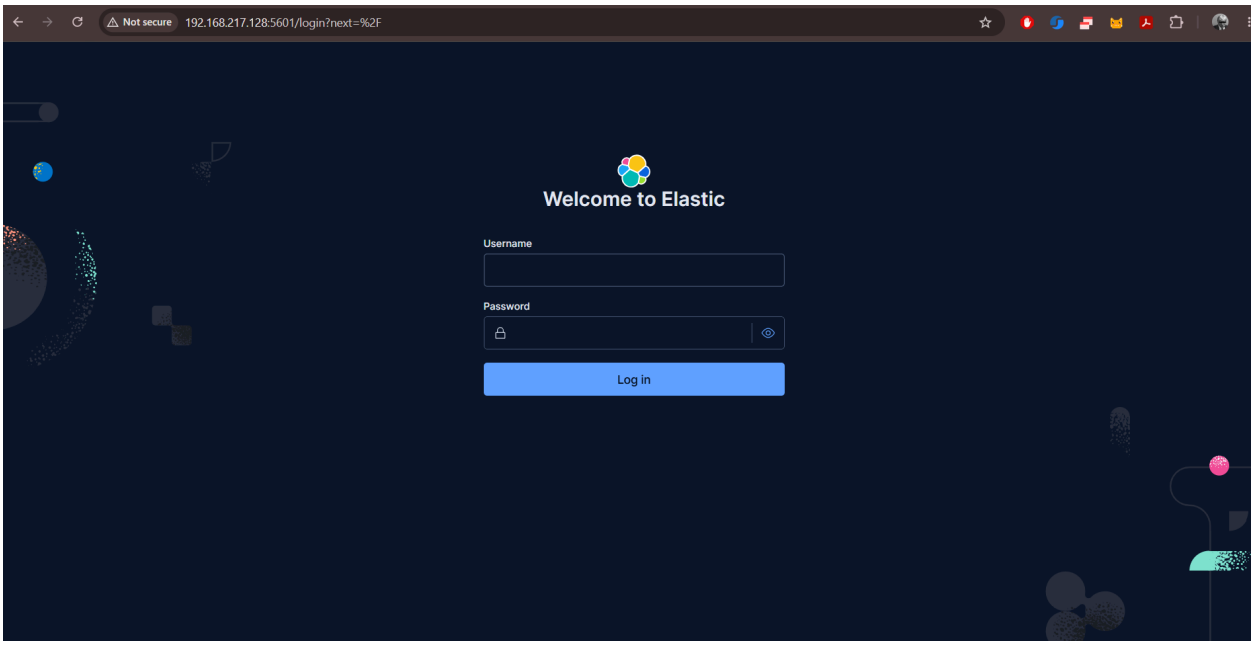
```
sudo nano /etc/kibana/kibana.yml
```

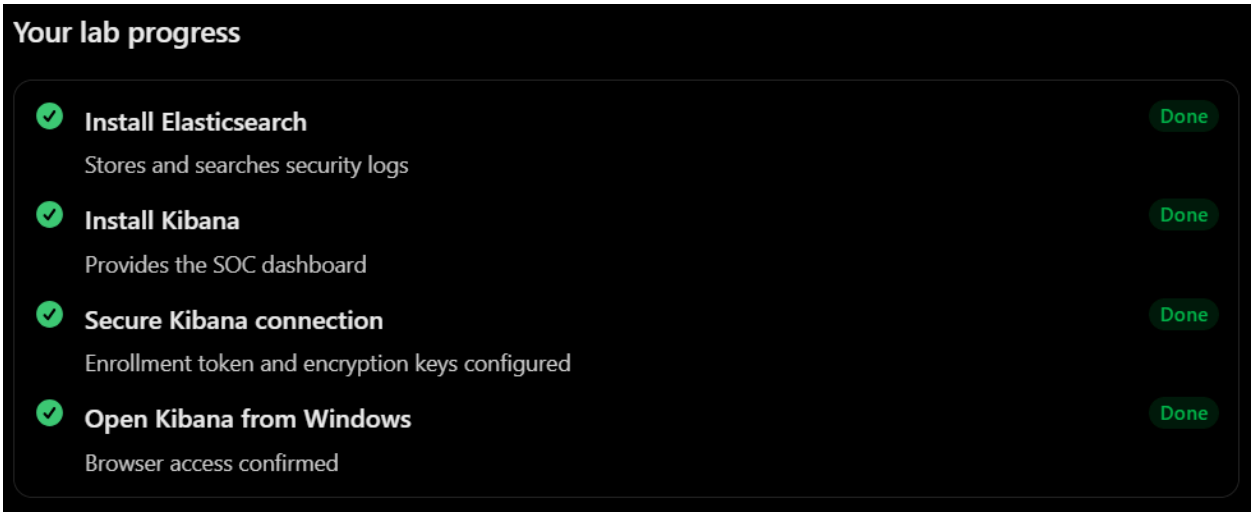
Go to the bottom and add the three generated entries

Step 7 — Open Kibana from Windows 11

```
Ubuntu Elastic SOC server
|
| Kibana listening on 0.0.0.0:5601
v
VMware private network
|
v
Windows 11 browser
|
v
Kibana login dashboard
```

```
http://YOUR-UBUNTU-IP:5601
```

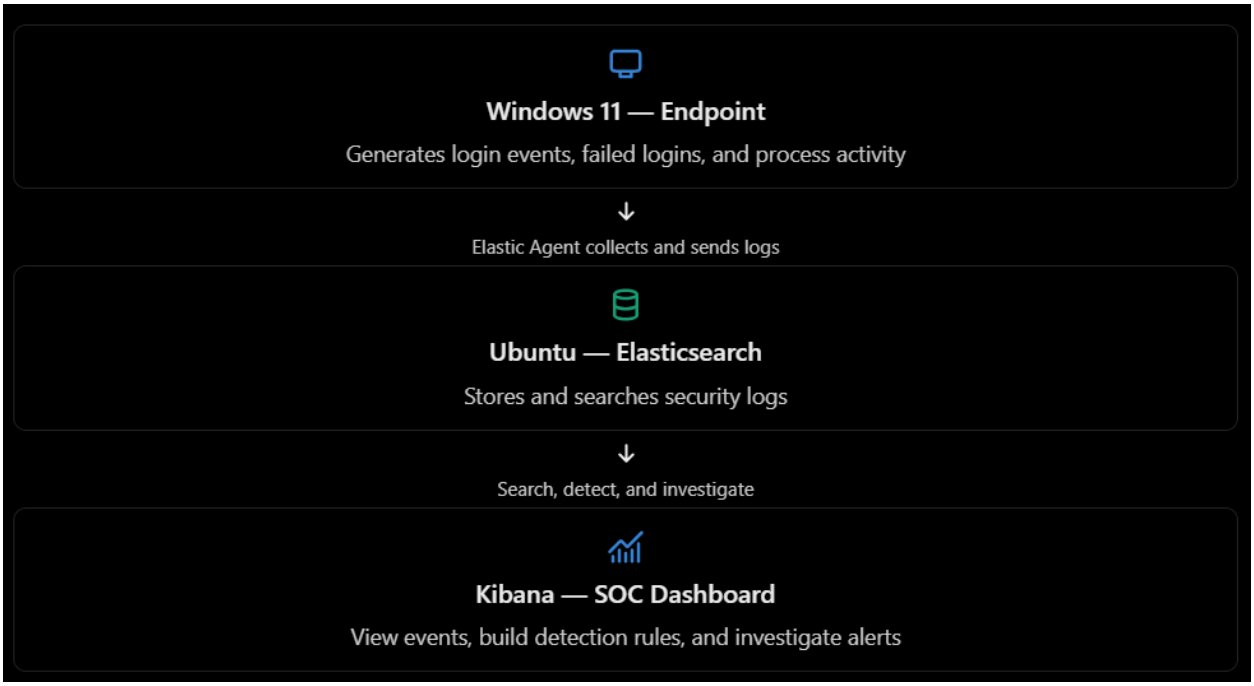




Stage 4 — Collect Windows security logs

```
Windows 11 computer
|
| Windows Event Logs
v
Elastic Agent
```

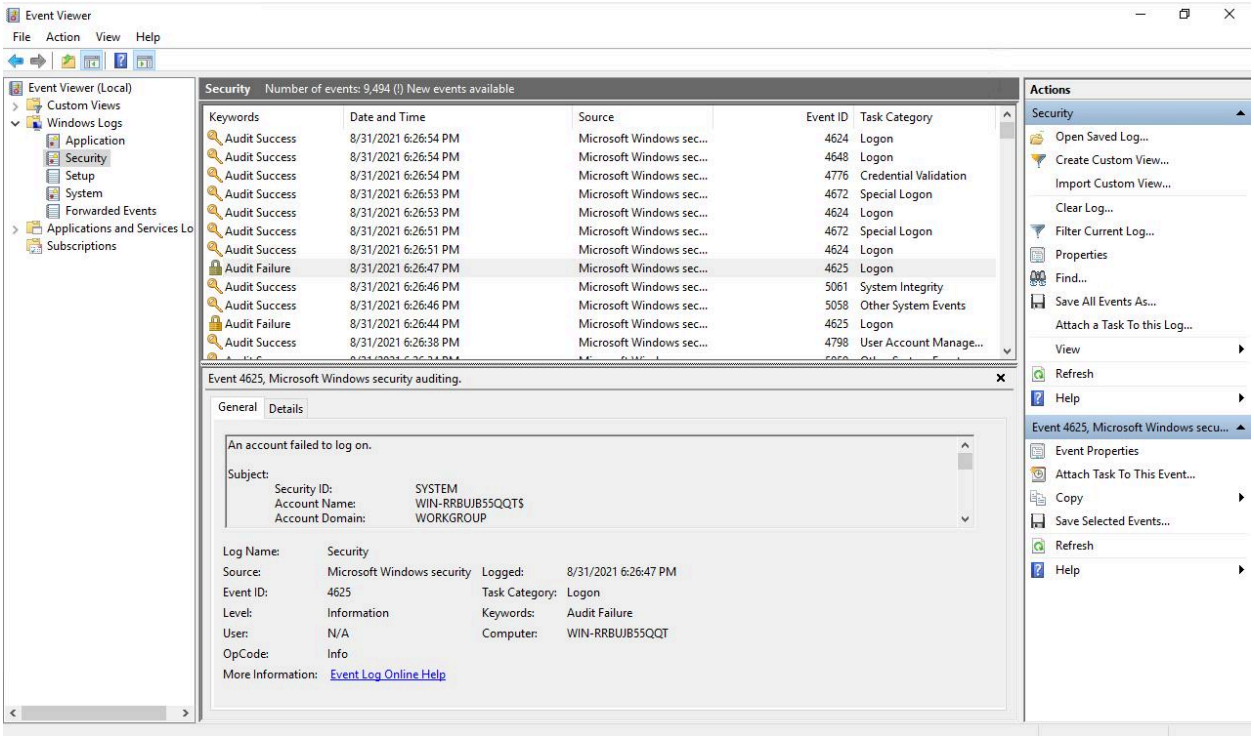
```
|
| Collect and send logs
v
Elasticsearch
|
| Search and analyze
v
Kibana → SOC investigation
```



What you will practice with Windows 11

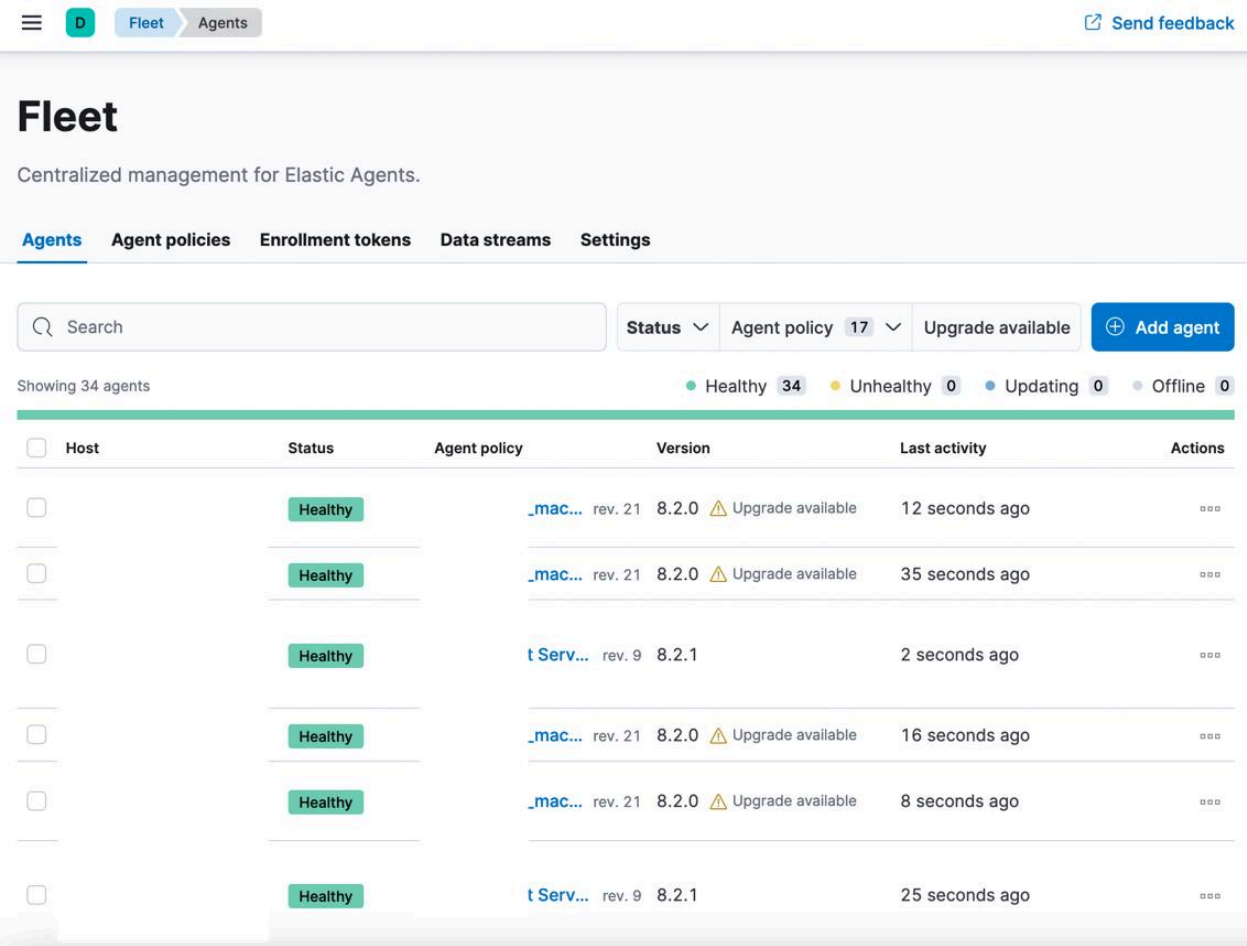
Practice 1 — Windows Event Logs

Learn to identify successful and failed logins, account activity, and other security events using Event Viewer.



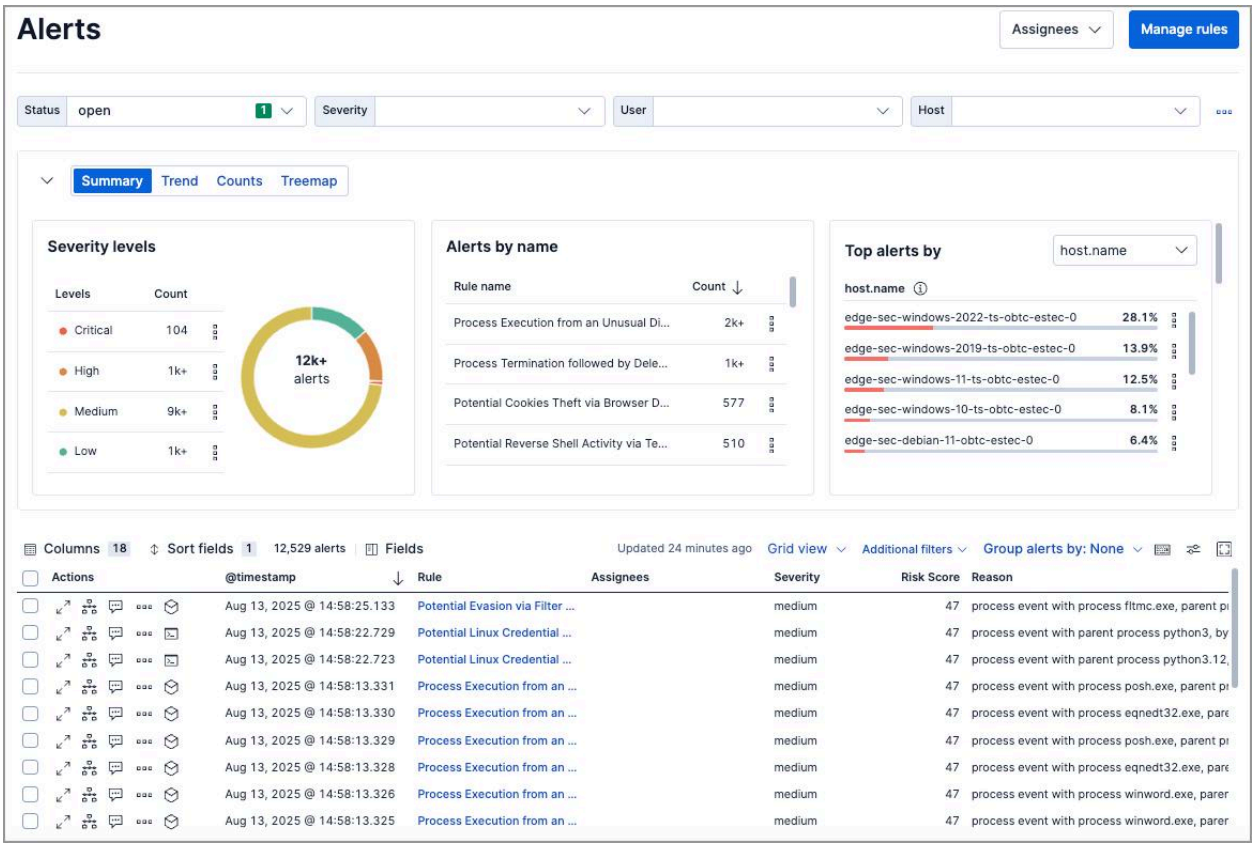
Practice 2 — Install Elastic Agent

Connect Windows to your Ubuntu server through Fleet and collect Windows security events.

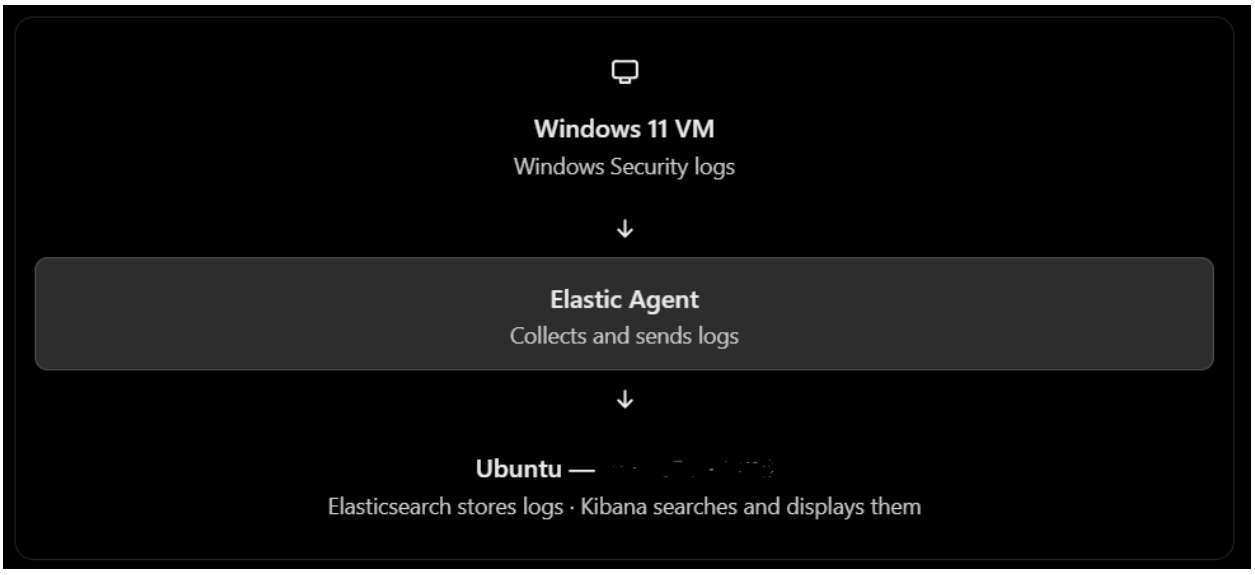


Practice 3 — Investigate security alerts

Search logs, investigate suspicious login activity, distinguish false positives from potential incidents, and document your findings.

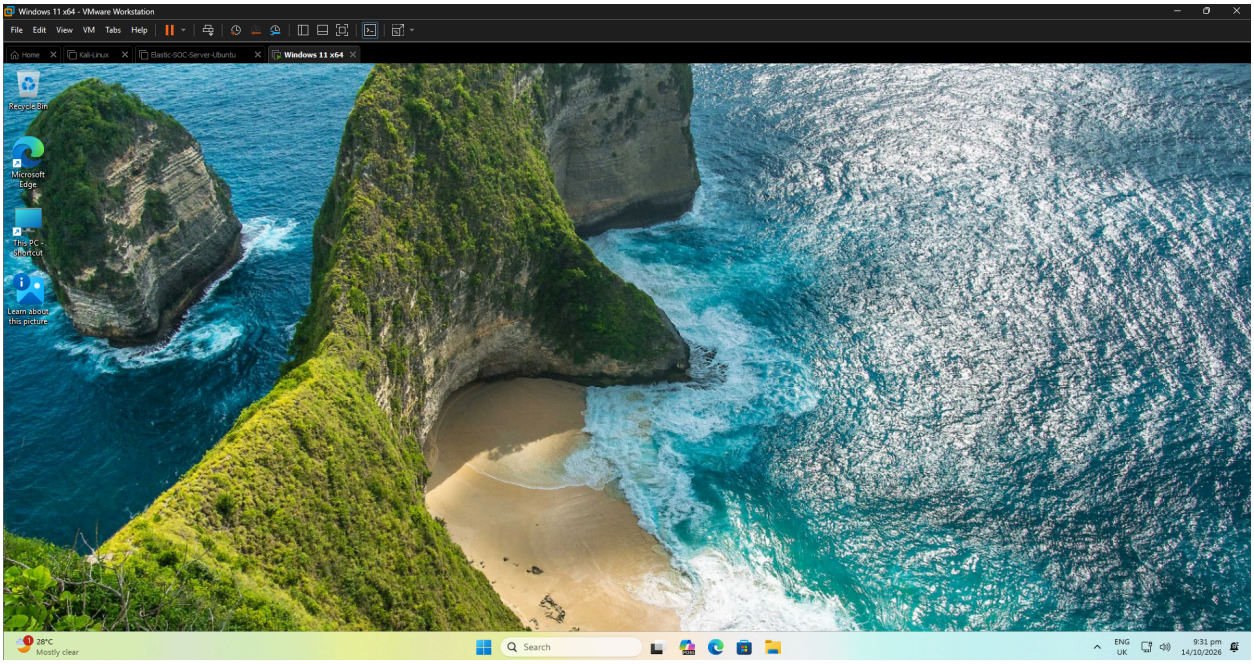


Stage 5 — Lab architecture

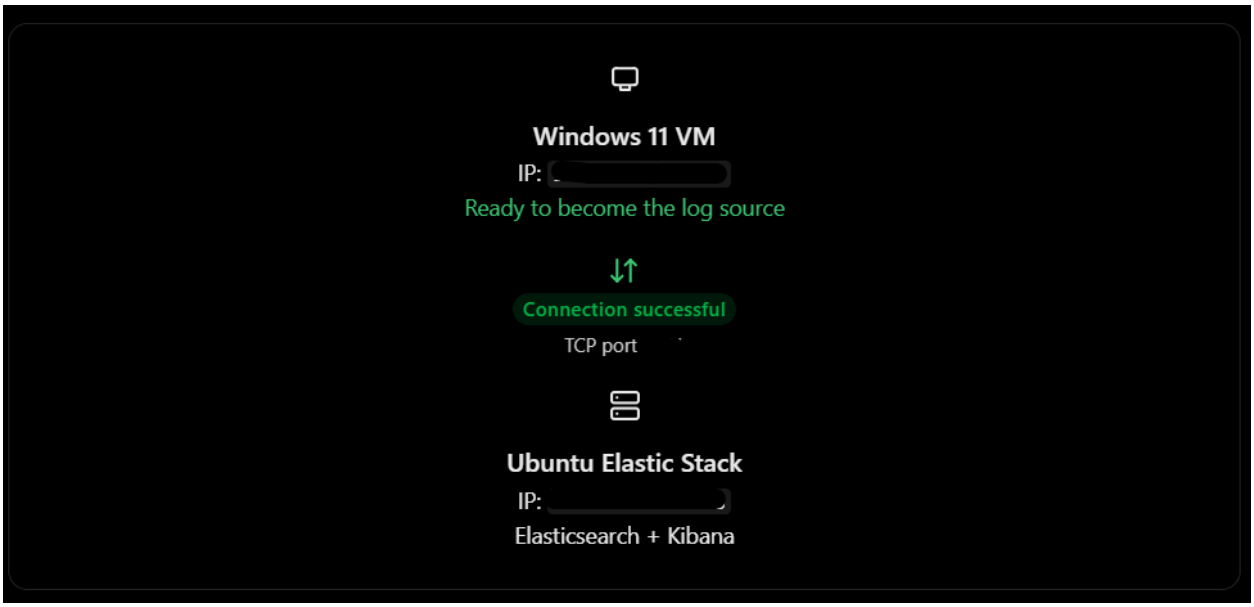


Step 1 — Create your Windows 11 VM

VMware setting	Recommended value
Guest OS	Windows 11 ×64 (Intel/AMD)
Memory	4 GB (4096 MB)
Processors	2 CPU cores
Virtual disk	64 GB, dynamically allocated
Network adapter	NAT
ISO	Your downloaded Windows 11 ISO



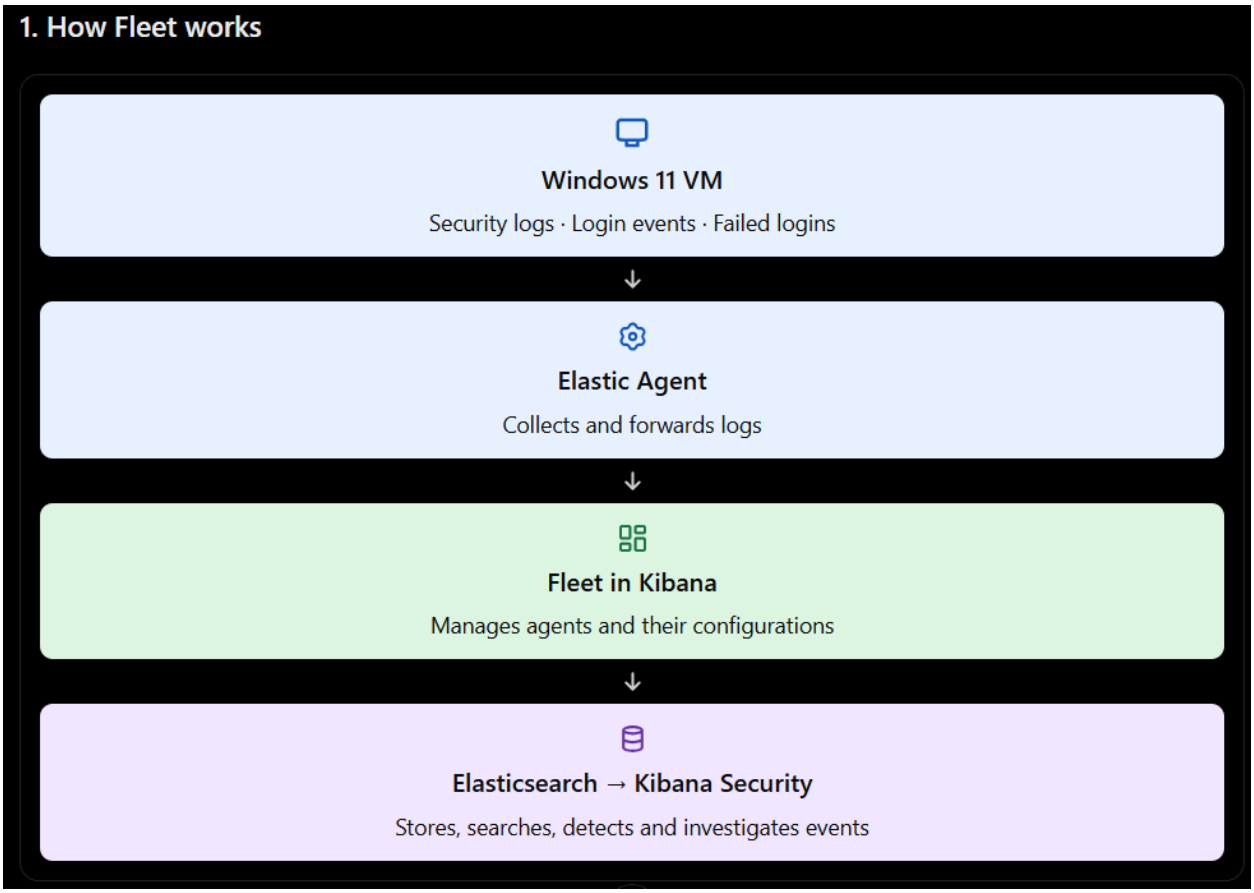
Step 2 — Start both VMs and check the network



Stage 6 — Configure Fleet Server

Elastic Fleet is a feature in Kibana that lets you centrally manage Elastic Agents installed on computers and servers. In your SOC Analyst L1 lab, Fleet helps you collect Windows security logs and send them to Elasticsearch for monitoring and threat detection.

Think of Fleet as the central control panel for the computers you want to monitor.



What are Fleet and Fleet Server?

Component	Purpose
Fleet	The management interface inside Kibana
Fleet Server	Coordinates communication between Fleet and enrolled Elastic Agents
Elastic Agent	Runs on your Windows VM and collects logs
Elasticsearch	Stores and indexes the collected logs
Kibana	Lets you search logs, build dashboards and investigate alerts

We'll do this in order:

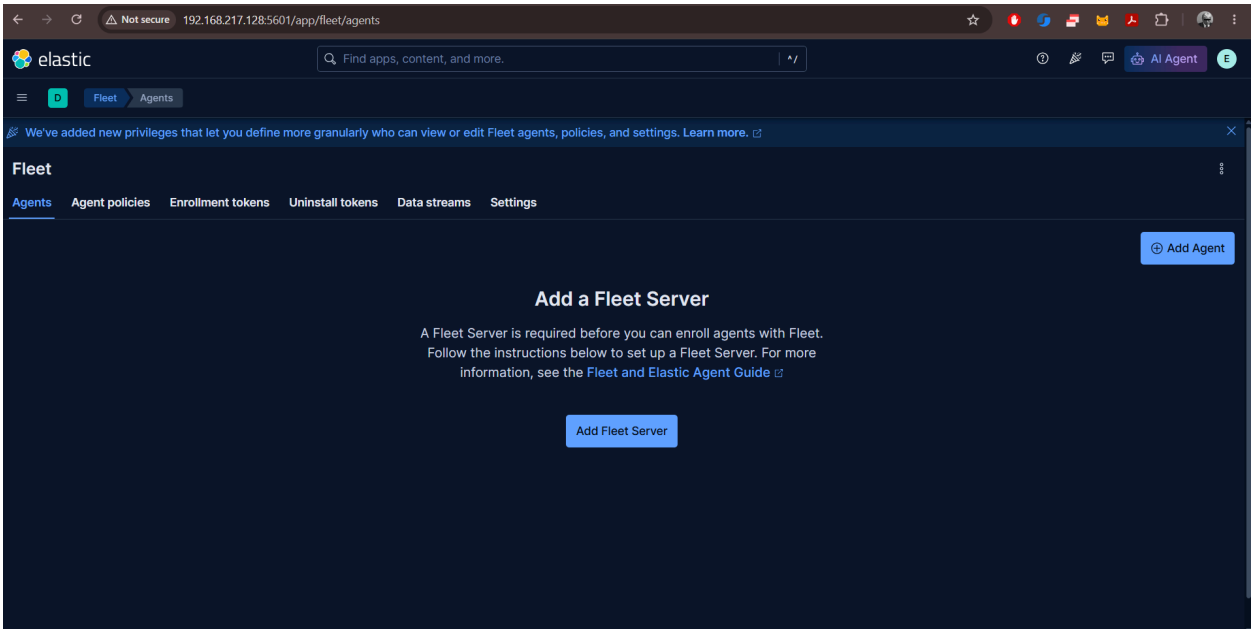
1. Configure Fleet Server on Ubuntu.
2. Confirm Windows can reach Fleet Server.
3. Install Elastic Agent on Windows.
4. Enable Windows Security event collection.

5. Verify the events in Kibana Discover.

Step 1 — Open Fleet

Then navigate to ☰ Menu → Management → Fleet.

Look for the Agents tab and an option such as Add Fleet Server.



Step 2 — Prepare Fleet Server in Kibana

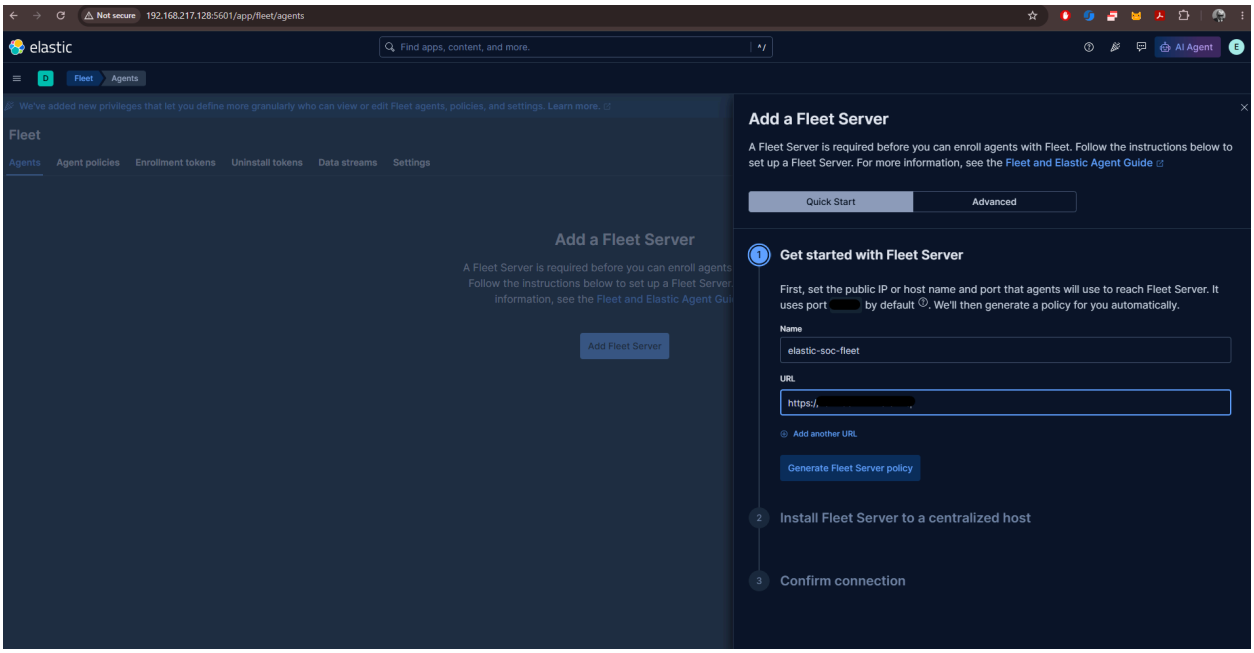
Reference: Elastic's self-managed 9.5.5 installation guide.

Elastic Docs

In Kibana, click Add Fleet Server.

1. Choose Quick Start, if that tab is available.
2. Set the Fleet Server name to `elastic-soc-fleet`.
3. Set the Fleet Server host URL to `https://localhost`.
4. Click Generate Fleet Server policy.

5. Select the Linux x86_64 installation instructions.



The URL uses your Ubuntu VM's IP address and Fleet Server's default port, `8220`. Quick Start uses a self-signed TLS certificate for this lab

Step 3 — Install Fleet Server on Ubuntu

Open the Ubuntu terminal. Run the commands that Kibana generates, one at a time, in Ubuntu.

The commands should download and install Elastic Agent with the Fleet Server role. Approve the installation if prompted, and wait for Kibana to confirm that Fleet Server has connected.

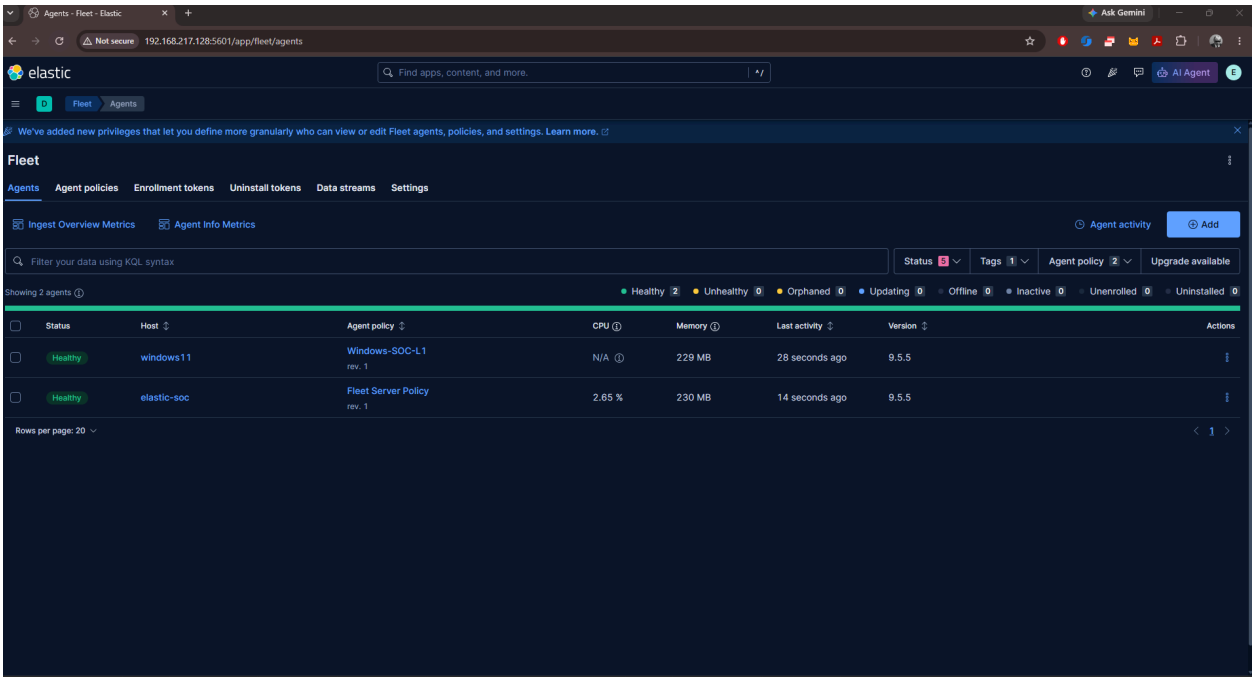
Step 4 — Open Fleet in Kibana

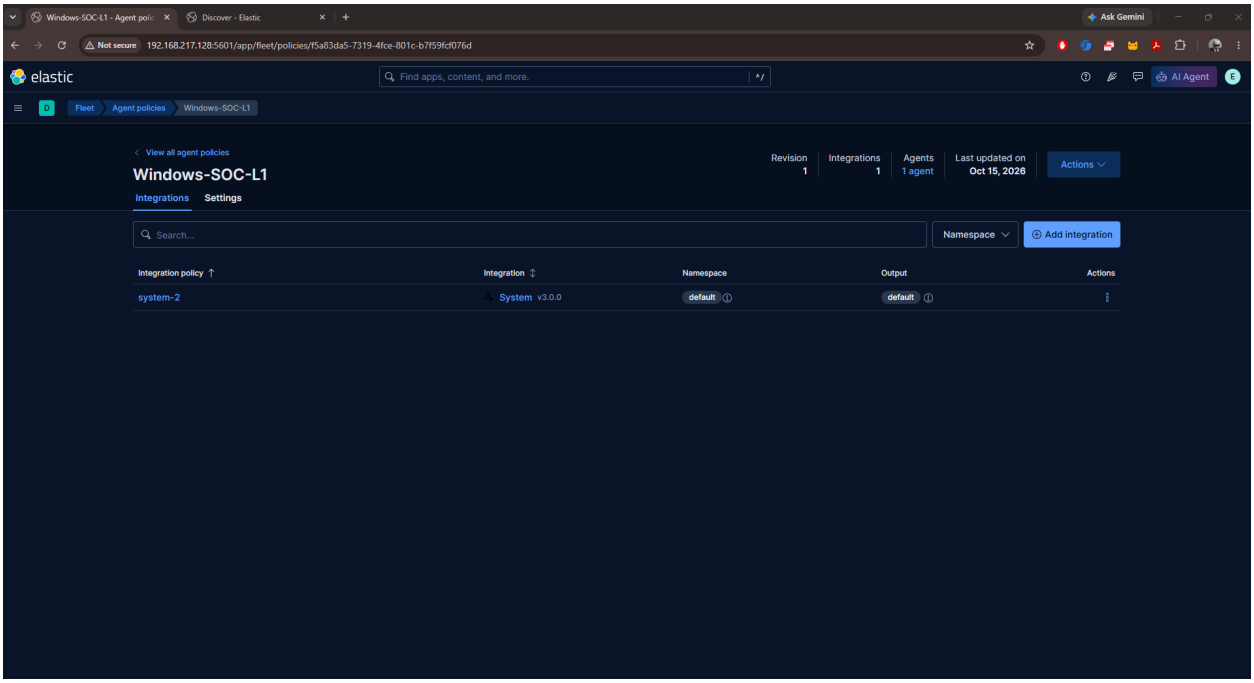
Then navigate to:

1. Management → Fleet

2. Open Agents
3. Click Add agent
4. Select or create an agent policy for your Windows VM.
5. Select Windows and the appropriate x86_64 / AMD64 installer option.

```
Elastic Agent has been successfully installed.
PS C:\WINDOWS\system32\elastic-agent-9.5.5-windows-x86_64>
```

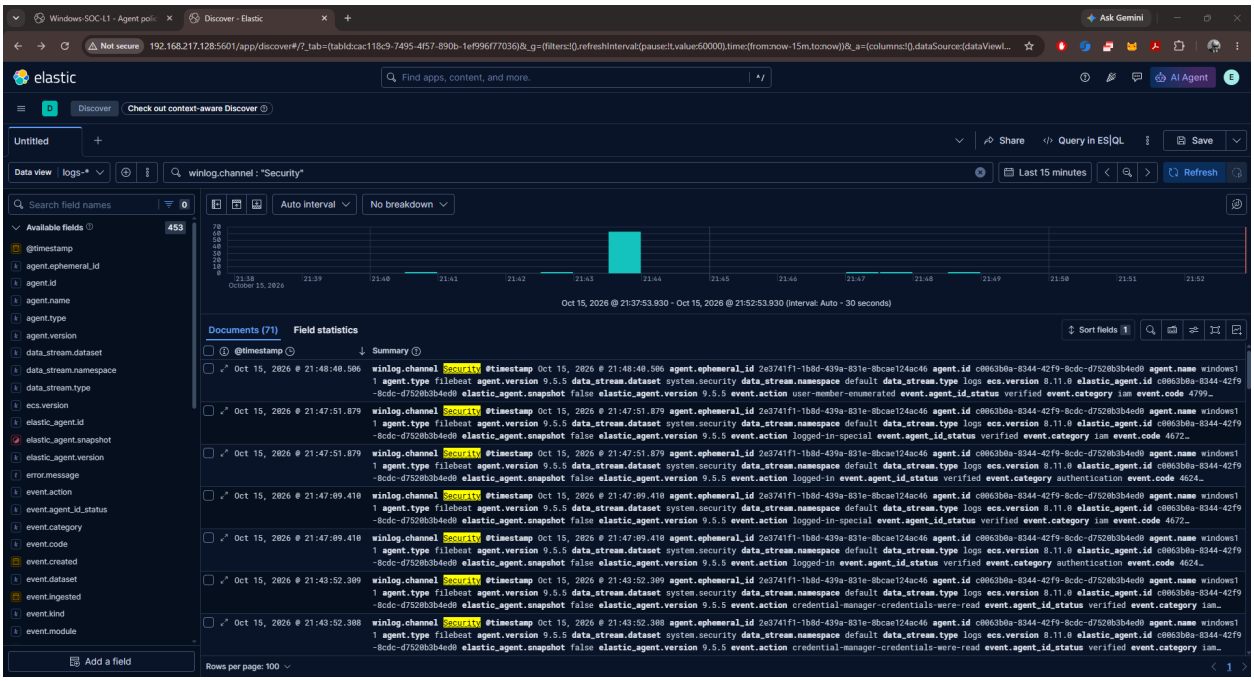


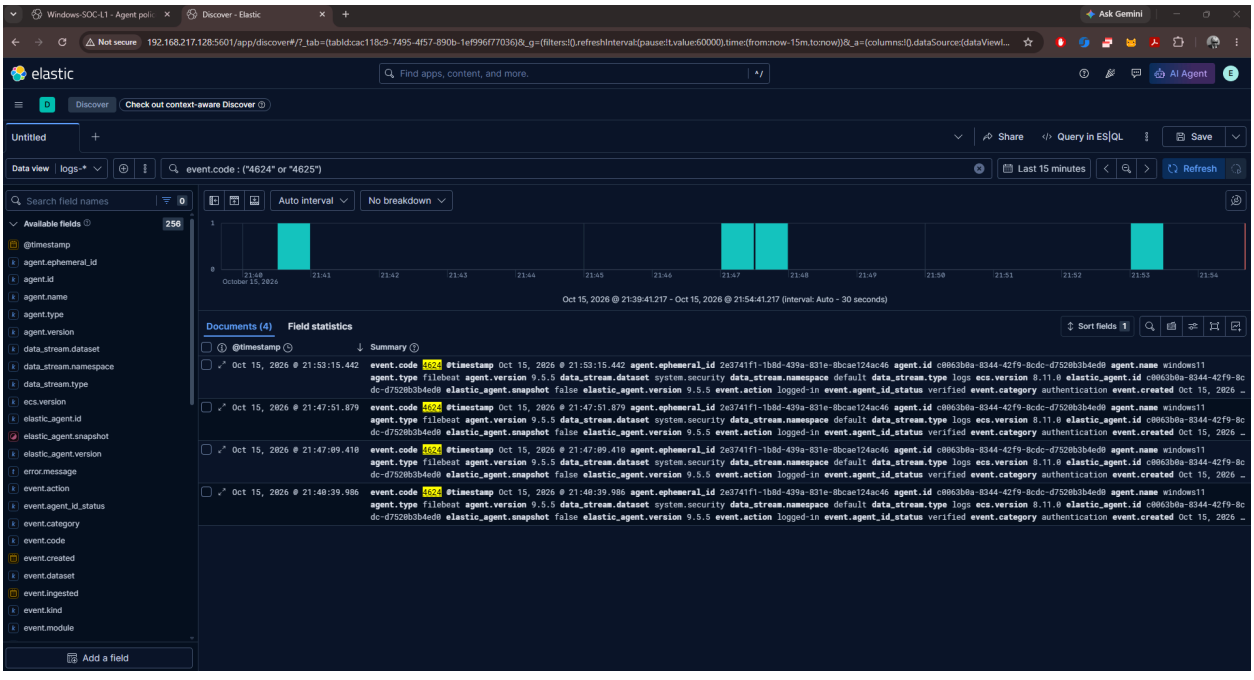


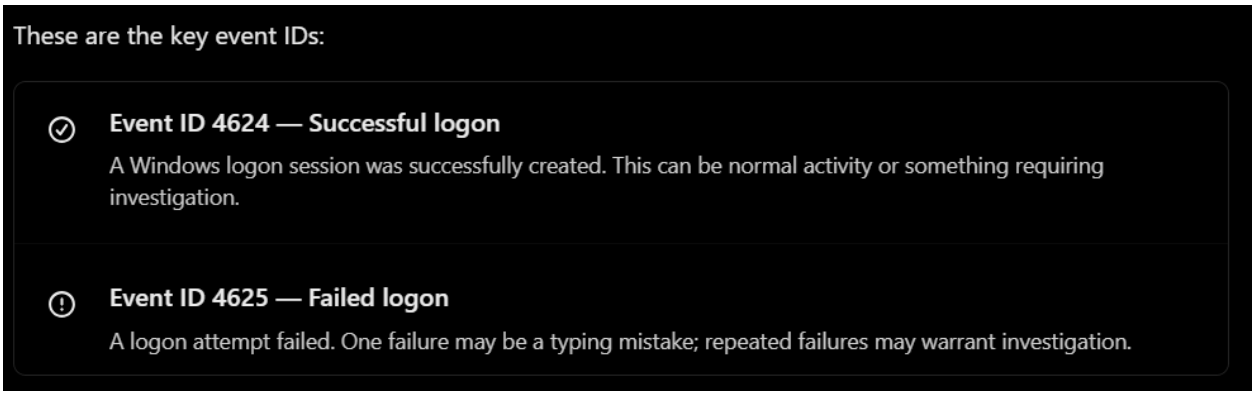
Step 1 — Open Discover

1. Open Kibana
2. Go to Analytics → Discover.
3. Look for a data view named `logs-system-*` or `logs-*`.
4. Set the time range to Last 15 minutes.

Step 5 — Search for Windows Security events


Windows-SOC-L1 - Agent polic
Discover - Elastic
Ask Gemini
Not secure
192.168.217.128:5601/app/discover#/?_tab=(tabId:cac118c9-7495-4f57-890b-1ef996f77036)&_g=(filters:!(),refreshInterval:(pause:!t,value:60000),time:(from:now-15m,to:now))&_a=(columns:!(),dataSource:(dataViewI...
elastic
Find apps, content, and more.
AI Agent
Discover
Check out context-aware Discover
Untitled
Share
Query in ES|QL
Save
Data view
logs-*
winlog.channel : "Security"
Last 15 minutes
Refresh
Search field names
Available fields
453
@timestamp
agent.ephemeral_id
agent.id
agent.name
agent.type
agent.version
data_stream.dataset
data_stream.namespace
data_stream.type
ecs.version
elastic_agent.id
elastic_agent.snapshot
elastic_agent.version
error.message
event.action
event.agent_id_status
event.category
event.code
event.created
event.dataset
event.ingested
event.kind
event.module
Add a field
Auto interval
No breakdown
Oct 15, 2026 @ 21:37:53.930 - Oct 15, 2026 @ 21:52:53.930 (interval: Auto - 30 seconds)
Documents (71)
Field statistics
Sort fields
@timestamp
Summary
Rows per page: 100


Windows-SOC-L1 - Agent polic
Discover - Elastic
Ask Gemini
Not secure
192.168.217.128:5601/app/discover#/?_tab=(tabId:cac118c9-7495-4f57-890b-1ef996f77036)&_g=(filters:!(),refreshInterval:(pause:!t,value:60000),time:(from:now-15m,to:now))&_a=(columns:!(),dataSource:(dataViewI...
elastic
Find apps, content, and more.
AI Agent
Discover
Check out context-aware Discover
Untitled
Share
Query in ES|QL
Save
Data view
logs-*
event.code : ("4624" or "4625")
Last 15 minutes
Refresh
Search field names
Available fields
256
@timestamp
agent.ephemeral_id
agent.id
agent.name
agent.type
agent.version
data_stream.dataset
data_stream.namespace
data_stream.type
ecs.version
elastic_agent.id
elastic_agent.snapshot
elastic_agent.version
error.message
event.action
event.agent_id_status
event.category
event.code
event.created
event.dataset
event.ingested
event.kind
event.module
Add a field
Auto interval
No breakdown
Oct 15, 2026 @ 21:39:41.217 - Oct 15, 2026 @ 21:54:41.217 (interval: Auto - 30 seconds)
Documents (4)
Field statistics
Sort fields
@timestamp
Summary



Enter this KQL query:

KQL query

```
event.code : ("4624" or "4625")
```